

An illustration of a young woman with long dark hair, wearing a blue shirt, sitting at a desk. She is looking at a laptop screen which displays a dashboard with various icons and charts. On the desk, there are several notebooks, pens, and a small container with more pens. One of the notebooks is open, showing handwritten notes in blue ink. The background is a solid light purple color.

AI-Powered Study Buddy

Capstone project — An intelligent learning assistant that explains concepts, summarizes notes, and generates quizzes & flashcards instantly.

CS & ENGINEERING

SAHARSA COLLEGE OF ENGINEERING

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Outline

Problem

Learning friction: scattered info, limited guidance, slow note-making.

Solution

An AI assistant for explanations, summaries, quizzes, and flashcards.

Design

Frontend, backend, AI engine, and data store.

Results

Faster understanding, efficient revision, improved prep.

Presentation flow: problem → solution → architecture → implementation → evaluation → next steps.

Problem Statement

Students face information overload and uneven access to guidance.

Common challenges:

1

Search overload

Search results are long, unfocused, and time-consuming.

2

Limited instructor access

Instructors are not always available for immediate help.

3

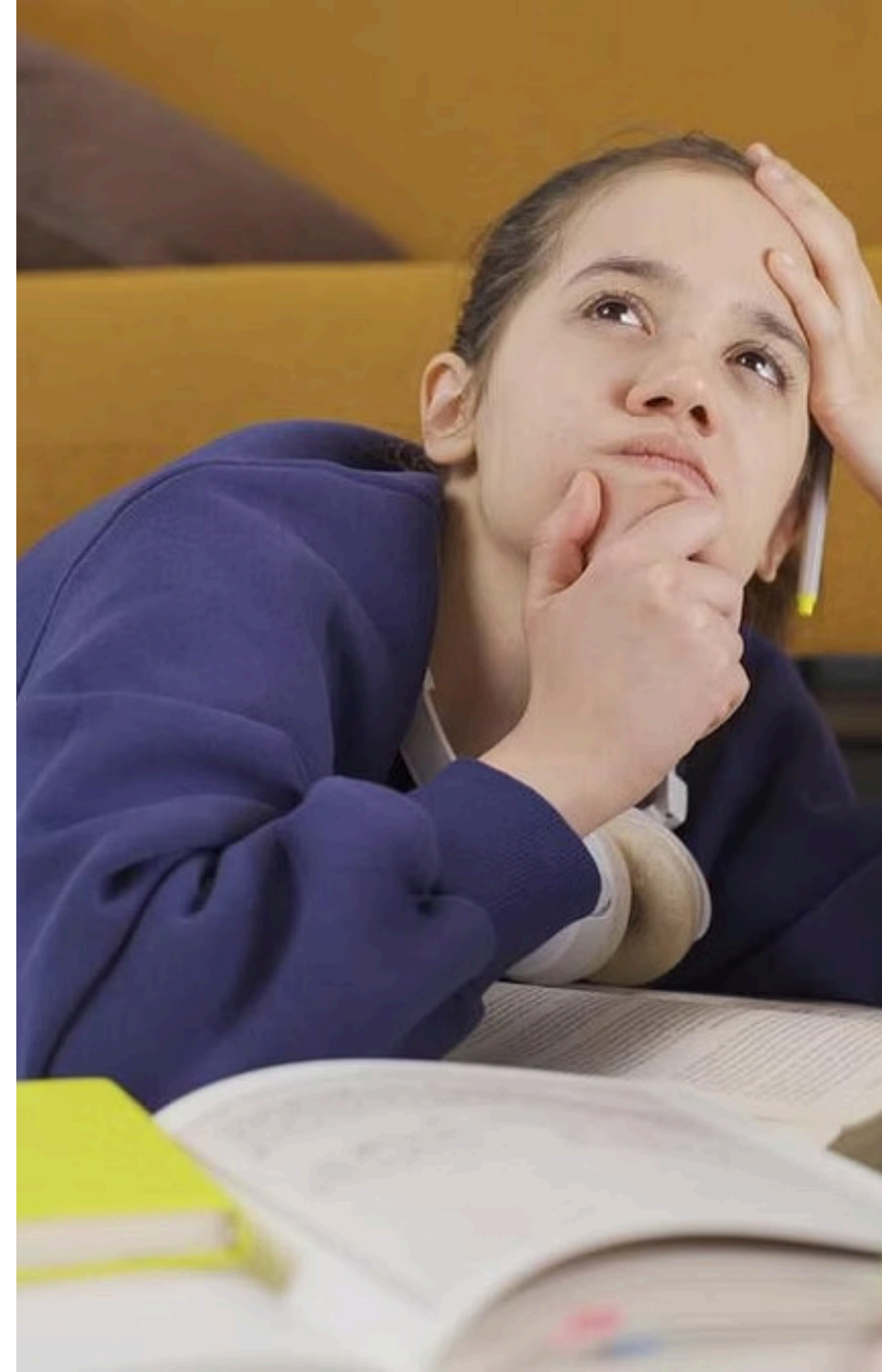
Slow note-making

Preparing concise notes and revision materials is tedious.

4

Need for instant support

Students need fast, personalized academic support during study sessions.



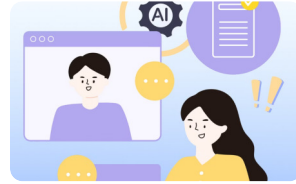
Proposed Solution

The Study Buddy is an AI-driven assistant that accepts student queries and returns:



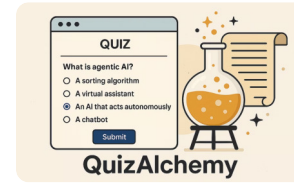
Concept explanations

in simple language



Automatic summarization

of long notes



Generated quizzes

(MCQs & practice problems)



Flashcards

for rapid revision

 Personalization adapts difficulty and style based on student needs and prior interactions.

Objectives & Key Features



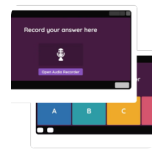
Explain Concepts

Translate complex topics into clear, concise explanations with examples.



Summarize Notes

Condense lengthy materials into structured, study-ready summaries.



Quiz Generator

Produce MCQs and practice questions with answer keys for self-assessment.



Flashcards

Create concise flashcards for spaced repetition and quick revisions.

System Architecture



This pipeline routes queries through the frontend to the AI engine, consults the knowledge base, and returns structured learning artifacts in real time.

Technology Stack

- Programming: Python
- Frontend: Streamlit, HTML, CSS
- AI: Hugging Face Transformers, LLMs
- NLP Libraries: NLTK, spaCy
- Database: SQLite (prototype), option for Firebase
- Deployment: Streamlit Cloud

❏ Stack chosen for rapid prototyping, reproducibility, and easy deployment to students.



Algorithm & Deployment Pipeline

01

1. Receive Query

Accept text input (question, topic, or notes).

02

2. NLP Analysis

Parse intent, extract key concepts, detect task type (explain/summary/quiz/flashcards).

03

3. Model Processing

Route to specialized LLM prompts or summarization pipelines (Hugging Face models).

04

4. Post-process & Format

Generate structured output: plain text explanation, concise summary, MCQs with answers, or CSV flashcards.

05

5. Deliver & Persist

Display in Streamlit UI and optionally save to SQLite/Firebase for user history and personalization.

Deployment: containerized Python app → Streamlit Cloud for student access; model hosting via Hugging Face or cloud inference endpoints.

Results, Conclusion & Future Work

- Results: Faster comprehension and reduced study time; effective revision with generated flashcards; improved self-assessment through quizzes.
- Conclusion: Study Buddy is a practical AI aid that augments student learning and study efficiency.
- Future Scope: Add voice tutor, multi-language support, mobile app, adaptive study plans, performance analytics, LMS integration.

✅ Next steps: prototype evaluation with students, collect usage data, refine personalization models.



References & Acknowledgements

Key resources and tools used during development:

- [Python Documentation](#)
- [Hugging Face](#)
- [Streamlit Documentation](#)
- [Scikit-Learn](#)

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GitHub Link :- <https://github.com/anchal-03/AI-Powered-Study-Buddy.git>

THANK YOU

